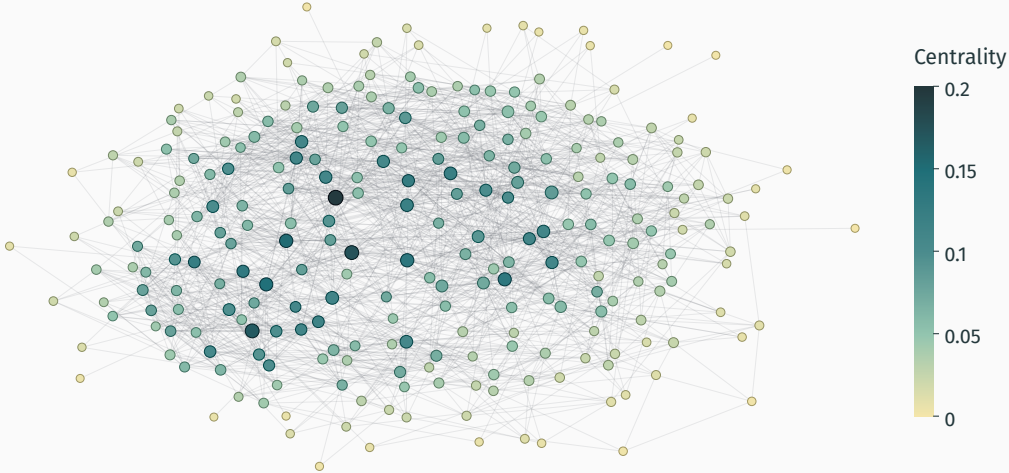
Centrality Estimation from Sampled Networks

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Network and centrality



Village 45 (Banerjee et al. 2024), largest connected component. Size: degree. Color: eigenvector centrality.

Use of centrality

1. Mechanism: downstream regression

How does outcome Y_i relate to household i's centrality v_i?

$$Y_i = \beta\, v_i(G) + Z_i^\top \theta + \epsilon_i, \qquad i \in V,$$

with controls Z_i and regression error ϵ_i.

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2. Policy: whom to seed?

- *Simple diffusion*: a few more random seeds suffice.
- Complex diffusion: central seeds must be found.
  - Information, e.g. micro-finance, vaccines
  - Technology, e.g. pit planting, hybrid seeds

Akbarpour et al. (2025), Banerjee et al. (2019), Beaman et al. (2021).

Question

How to consistently estimate centrality from a sampled network?

Network survey is costly

Cost component ($)	Total	Per village
Census	29,904	249
Network survey	84,954	708
Data entry and matching	14,284	119
Tablet rentals	8,584	72
Project staff salaries	20,185	168
Travel	1,617	13
Training and meetings	1,916	16
Office expenses	3,047	25
Overhead (15%)	24,674	206
Total cost	**189,164**	**1,576**
Duration	**8.2 months**	

Source: Breza et al. (2020), Table 4. J-PAL South Asia, full census and network survey of 120 villages.

The classical remedy, and why it fails

Interview a subset of households, $S \subsetneq V$.

Benchmark: simple random sampling

Draw $|S|$ households from V uniformly at random without replacement.

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Why it fails: centrality is global

Every missing link enters everyone's centrality,

$$v \propto \lim_{k \to \infty} \lambda^{-k} A^k \mathbf{1}.$$

Literature

Bias correction

Chandrasekhar & Lewis (2016)

- **Target.** Downstream slope β.
- **Data.** Simple random sample.
- **Method.** Analytical correction.
- No correction for spectral statistics such as centrality.
- Non-classical measurement error: $\hat{\beta}$ attenuates, expands, or flips sign.

Bound

Thirkettle (2019)

- **Target.** Centrality statistics.
- **Data.** Simple random sample.
- **Method.** Strategic network-formation model.
- Set identified (missing links, multiple equilibria), so bounds only.

Model

Breza et al. (2020)

- **Target.** Node- and graph-level statistics.
- **Data.** Aggregated relational data (ARD).
- **Method.** Fit model $\rightarrow$ simulate graphs.
- Parametric model with conditionally independent links.
- No actual link observed.

This paper: conditioning on the network

- **Network formation is never modeled**
  - No DGP misspecification
  - No equilibrium selection
  - No covariates required
- **Allows network externalities**
  - No assumption on (conditional) independent links
- **Allows network sparsity and hubs**

Contributions

- **Estimation.** Propose a sequential sampling design with a consistent centrality estimator
- **Inference.** At the null, an exactly unbiased slope and an asymptotically exact robust t-test. Away from it, a bootstrap bias correction
- **Evidence.** Surveying half the households recovers full-network slopes within 1.1%. Simple random sampling understates them by 11–12%

Sequential sampling protocol

(i) Start. Interview a uniformly random household and record all its contacts.

(ii) Draw. Pick a uniformly random contact of the current household. It becomes the current household.

(iii) Interview if new. Interview the current household if it has not been interviewed before. In both cases, the next household is drawn from its recorded contacts.

Repeat (ii)–(iii) until a sampling fraction α of households is interviewed.

Sampled network and estimator

Definition 1 (Sample adjacency matrix)

The sample adjacency matrix of S is

$$\hat{A}_{ij} := A_{ij}\,\mathbf{1}\{i \in S \text{ or } j \in S\},$$

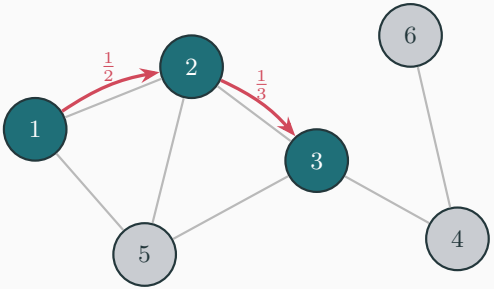
and $\hat{G}$ is the network with adjacency matrix $\hat{A}$.

Definition 2 (Estimator)

Let $\hat{\lambda} := \lambda_1(\hat{A})$. The estimated eigenvector centrality $\hat{v}$ is a nonnegative unit vector satisfying

$$\hat{A}\,\hat{v} = \hat{\lambda}\,\hat{v}.$$

Illustration



sampled $S = \{1, 2, 3\}$ unsampled $\bar{S} = \{4, 5, 6\}$

$$P = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ \frac{1}{3} & 0 & \frac{1}{3} & 0 & \frac{1}{3} & 0 \\ 0 & \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \end{array}$$

$$P := D^{-1}A, \qquad D = \mathrm{diag}(A\mathbf{1})$$

Analysis strategy

Step 1: Matrix perturbation

Deterministic, given the sample S

Split the adjacency matrix $A = \hat{A} + E$:

$$\hat{A} = \underbrace{\begin{pmatrix} A_{SS} & A_{S\bar{S}} \\ A_{\bar{S}S} & 0 \end{pmatrix}}_{\text{observed}}, \quad E = \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & A_{\bar{S}\bar{S}} \end{pmatrix}}_{\text{missing}}.$$

Question

- How far do the missing links in E move $\hat{v}$ away from v?

Step 2: Markov framework

Stochastic, over the distribution of S

Model the sampling process as a random walk on the graph G.

Questions

- Which households are unsampled?
- Which links end up missing?

Ingredients of the rate $r_n = \bar{d}/(\alpha\gamma\delta\lambda)$

α **Sampling fraction.** The budget parameter chosen by the researcher.

$\bar{d}$ **Average degree.** $\bar{d} = \frac{1}{n}\sum_{i\in V} d_i$.

λ **Leading eigenvalue.** $\lambda = \lambda_1(A) \geq \bar{d}$, with equality only on regular graphs.

δ **Relative spectral gap.** $\delta := 1 - \lambda_2(A)/\lambda$

γ **Mixing speed.** $\gamma := 1 - \lambda_2(P)$, where $P = D^{-1}A$.

d_h **Harmonic mean degree.** $d_h := \left(\frac{1}{n}\sum_{i\in V} d_i^{-1}\right)^{-1}$.

Main theorem

The estimator converges at the rate

$$r_n := \bar{d}/(\alpha\gamma\delta\lambda)$$

under two conditions on the graph sequence.

Assumption 1 (Network sequence)

(i) *Rate condition.* $r_n \log n \to 0$ as $n \to \infty$.

(ii) *Degree-balance condition.* $\limsup_{n\to\infty} \bar{d}/d_h < \infty$.

Theorem 1 (Consistency and rate of convergence)

Let G_n be connected graphs on n nodes, each sampled by the sequential sampling design at fraction $\alpha \in (0, 1]$. Under Assumption 1,

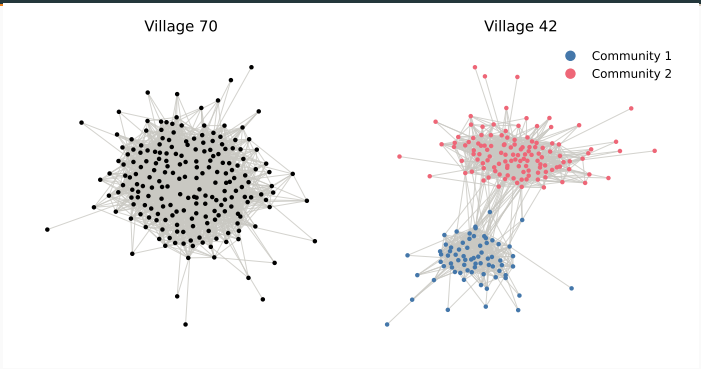
$$\left\|\hat{v} - v\right\|_2 \;=\; O_p(r_n).$$

Implications for empirical research

Which networks are compatible

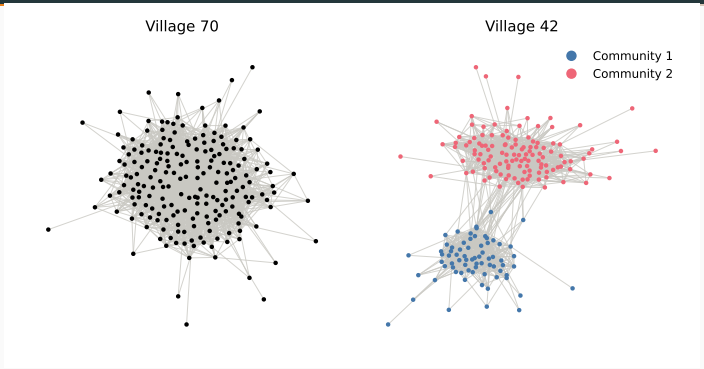
- **Degree heterogeneity** $\lambda \gg \bar{d}$
  Some households have far more links than average.
- **Degree balance** $\bar{d}/d_h$ bounded
  Few near-isolated households.
- **No bottleneck** γ not too small
  Sampling spreads across the village.

Community structure in real villages



Village 70: no bottleneck ($\gamma = 0.422$). Village 42: two communities, few bridges ($\gamma = 0.045$).

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- **Bottleneck: sample more, or start a second surveyor at random in $V \setminus S$.**

Diagnostic: draws keep returning to interviewed households while most remain unsampled.

Regression: full vs. sampled network

Full network

Given A, compute v.

$$Y_i = \beta\, v_i + Z_i^\top \theta + \underbrace{\epsilon_i}_{\text{uncorrelated with } v_i}$$

Sampled network

Given $\hat{A}$, compute $\hat{v}$.

$$Y_i = \beta\, \hat{v}_i + Z_i^\top \theta + \underbrace{\epsilon_i + \beta(v_i - \hat{v}_i)}_{\text{correlated with } \hat{v}_i}$$

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Assumption 2 (Identifying variation)

(i) Z has full column rank.

(ii) $\frac{1}{n}\, v^\top M_Z v \geq \mu_0 > 0$ for all large n, $\quad M_Z := I - Z(Z^\top Z)^{-1} Z^\top$.

Slope consistency

Assumption 3 (Regression error: consistency)

(i) *Exogenous sampling.* Conditional on the network G, sampling is independent of the regression error ϵ and the controls Z: $S \perp (\epsilon, Z) \mid G$.

(ii) *Conditional mean zero.* $\mathbb{E}[\epsilon \mid G, Z] = 0$.

(iii) *Bounded second moments.* Conditional on (G, Z), the errors are uncorrelated with $\mathrm{Var}(\epsilon_i \mid G, Z) \leq \bar{\sigma}^2$ for all $i \in V$, for a constant $\bar{\sigma}^2 < \infty$.

Proposition 1 (Slope consistency)

Let the conditions of Theorem 1 hold together with Assumptions 2 and 3. Then, conditional on (G, Z),

$$\hat{\beta} \xrightarrow{p} \beta.$$

Inference at the null

Assumption 4 (Regression error: inference)

Conditional on (G, Z), the errors $\epsilon_1, \ldots, \epsilon_n$ are independent, with $\sigma_i^2 := \mathrm{Var}(\epsilon_i \mid G, Z) \geq \underline{\sigma}^2 > 0$ and $\mathbb{E}[\epsilon_i^4 \mid G, Z] \leq C < \infty$ for all $i \in V$.

Assumption 5 (Negligible leverage)

In the full-network regression, $\max_{i \in V} X_i^\top (X^\top X)^{-1} X_i \to 0$.

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Proposition 2 (The t-test at the null)

Let the conditions of Theorem 1 and Assumptions 2, 3, 4, and 5 hold. With $\hat{w} := M_Z \hat{v}$ and $\hat{\epsilon}$ the residuals of the feasible regression, define

$$t := \frac{\hat{\beta}}{\hat{\omega}}, \qquad \hat{\omega}^2 := \frac{\sum_{i \in V} \hat{w}_i^2 \, \hat{\epsilon}_i^2}{(\hat{w}^\top \hat{w})^2}.$$

Under $\mathcal{H}_0 : \beta = 0$, conditional on (G, Z), $t \xrightarrow{d} N(0, 1)$.

Simulation

	Sequential sampling			Simple random sampling		
Budget	10%	30%	50%	10%	30%	50%
Barabási–Albert						
Relative bias $\hat{\beta}/\beta - 1$	−0.294	−0.108	−0.045	−0.547	−0.269	−0.122
$\mathrm{SD}(\hat{\beta})$	0.069	0.044	0.045	0.101	0.063	0.050
Chung–Lu						
Relative bias $\hat{\beta}/\beta - 1$	−0.270	−0.089	−0.032	−0.497	−0.237	−0.105
$\mathrm{SD}(\hat{\beta})$	0.061	0.043	0.043	0.091	0.057	0.047
Erdős–Rényi						
Relative bias $\hat{\beta}/\beta - 1$	−0.894	−0.739	−0.532	−0.901	−0.755	−0.541
$\mathrm{SD}(\hat{\beta})$	0.048	0.064	0.081	0.048	0.067	0.087
Stochastic block model						
Relative bias $\hat{\beta}/\beta - 1$	−0.896	−0.743	−0.536	−0.903	−0.756	−0.542
$\mathrm{SD}(\hat{\beta})$	0.048	0.064	0.082	0.048	0.067	0.087

Notes: Means over 100 graphs and 20 sampling replications per design at $n = 1000$. The outcome model is $Y_i = \beta v_i + c + \epsilon_i$ with ϵ_i i.i.d. $N(0, 1)$ independent of the sampling process and $\beta = 1.5$. The relative bias row reports $\mathbb{E}[\hat{\beta} \mid S, G, Z]/\beta - 1$, computed exactly for each realized sample, whose mean over samples estimates $\bar{\rho} - 1$. The SD row reports, averaged over graphs, the standard deviation of $\hat{\beta}$ within a graph, which combines the dispersion of the conditional bias across samples with the mean conditional variance. Monte Carlo standard errors across graphs are below 0.002 for every entry. Budgets and sampling designs are as in the first-stage table.

Application: Karnataka village networks (Banerjee et al. 2019)

- Source: Wave 2 of the Karnataka village surveys Banerjee et al. (2019, 2024)
- Why centrality matters:
  - Nominated people are central
  - Increased monthly child immunizations by 22% vs. random seeds

33 gossip villages	Mean	SD
Network size n	222	68
Average degree $\bar{d}$	17.4	3.8
Maximum degree $d_{\max}$	53	17
Leading eigenvalue λ	22.8	5.0
Relative spectral gap δ	0.327	0.099
Mixing speed γ	0.223	0.099
Degree balance $\bar{d}/d_h$	1.70	0.24
Share nominated (loan)	0.048	0.029

Application: replicate full-network result under sampling

	Sequential sampling			Simple random sampling		
Budget	10%	30%	50%	10%	30%	50%
Loan nominations ($\beta_{\text{oracle}} = 0.378$)						
Relative bias $\hat{\beta}/\beta_{\text{oracle}} - 1$	−0.386	−0.110	−0.011	−0.528	−0.254	−0.114
$\text{SD}(\hat{\beta})$	0.057	0.034	0.018	0.053	0.048	0.032
Event nominations ($\beta_{\text{oracle}} = 0.283$)						
Relative bias $\hat{\beta}/\beta_{\text{oracle}} - 1$	−0.419	−0.110	−0.005	−0.522	−0.261	−0.121
$\text{SD}(\hat{\beta})$	0.044	0.028	0.017	0.050	0.033	0.026

Notes: Means over 100 sampling replications per design and budget. The regression is the unweighted pooled least squares of the nomination count on standardized centrality with a constant, on the $6{,}466$ estimation rows of the 33 gossip villages, the specification of Banerjee et al. (2019, Table C.1). β_{oracle} is the least-squares slope on the true centrality reported there, a fixed benchmark with no dispersion row, and is not the structural coefficient β. $\text{SD}(\hat{\beta})$ is the standard deviation of the feasible slope across sampling replications, in slope units. Centrality is the unit-norm eigenvector within each village, the plug-in centrality is computed on the sample adjacency matrix under the same normalization, and the pooled regressor is divided by its standard deviation over the estimation rows. Both designs interview $\lceil \alpha n_g \rceil$ households in village g of size n_g, isolated households included, and simple random sampling draws them without replacement, so the budget columns are comparable with the simulation tables.

- **Sequential sampling of half the households: within 1.1% of the full-network slope.**

Conclusion

- **Estimation.** A sequential sampling design estimates centrality consistently.
- **Inference at the null.** Exactly unbiased slope, asymptotically exact t-test.
- **Next.** A broader class of network statistics.

Thank you!